Operating System

What will be covered

- Defination of OS
- Functions of OS
- Types of OS
- Some concepts in OS
- Memory
- Kernel

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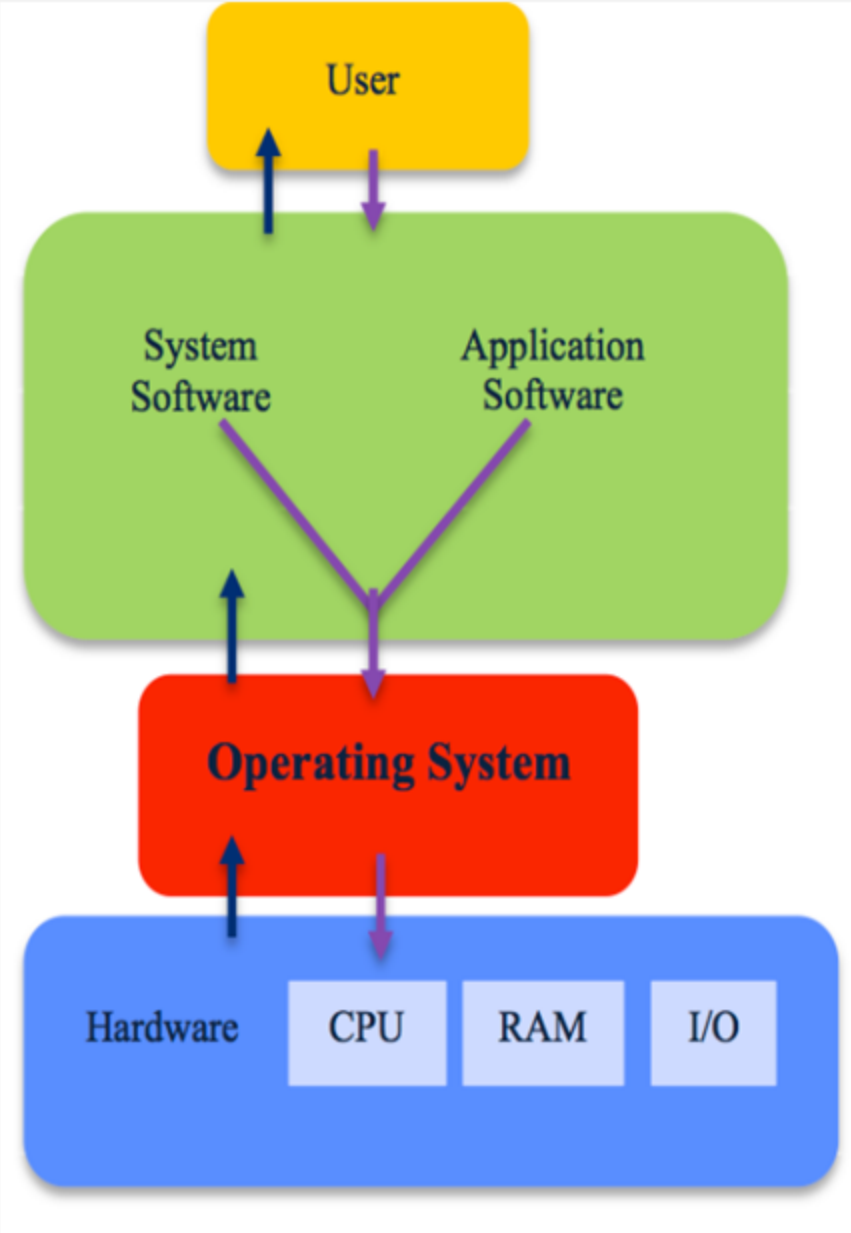
Defination of OS

Operating system is a...

- Interface between user an hardware.
- Resource allocator.
- Program controller.
- Command Interpreter

System S/w

- System software is a type of computer program that is designed to run computer's hardware and application programs.
- It is an interface between the hardware and user applications.



Examples of system software

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- **The BIOS (basic input/output system) ::**

  This system software ,gets the computer system started after it's turned on and manages the data flow between the OS and attached devices, such as the hard drive, video adapter, keyboard,mouse and printer.

- **The boot program ::**

  This system software loads the OS into the computer's main memory or random access memory (RAM).

- **An assembler ::**

  Takes basic computer instructions and converts them into a pattern of bits that the processor of computer can use to perform its basic operations.

- **A device driver ::**

  Controls a particular type of device that is attached to your computer, such as a keyboard or mouse.

Additionally, system software can also include system utilities like---

- ✓ Disk Defragmenter
- ✓ Compilers
- ✓ Debuggers.

Functions Of OS

Security : OS uses password protection to protect user data and similar other techniques to prevent unauthorized access to programs and user data.

Resourse Managment : OS ensures the proper use of all the available resources .

Process Managment : In a multi programming environment, the OS decides the order in which processes will have access to the processor, and how much processing time each process has. OS keeps track of the status of processes.

File Managment : A file system is organized into directories for efficient or easy navigation and usage.

Job accounting : Operating system Keeps track of time and resources used by various tasks and users.

Functions Of OS

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Error detection : Operating system constantly monitors the system to detect errors and avoid malfunctioning of computer system.

Memory Managment : The operating system manages the Primary Memory or Main Memory.

Coordination between S/Ws and users : Operating systems also coordinates and assigns interpreters, compilers, assemblers and other software to the various users of the computer systems.

Command Interpreter: Major functions of OS is to interprete the commands. OS provides an interface to the users for the communication with the Computer. Command interpreter is also known as a Shell. Command interprater provides a user interface to hide hardware details of the system from its users, making system more user friendly.

The two commonly used methods, provided by operating systems, for interfacing are

1. Command line Interface (CLI)
2. Graphical User Interface (GUI).

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Examples of Operating System

- Windows (GUI based, PC)
- GNU/Linux (Personal, Workstations)
- macOS (Macintosh), used for Apple's personal computers and work stations (MacBook, iMac).
- Android (Google's Operating System for smartphones/tablets/smartwatches)
- iOS (Apple's OS for iPhone, iPad and iPod Touch)

Popular OSs for computers are::

- ✓ Windows 10
- ✓ Mac OS X
- ✓ Ubuntu

Popular network/server OSs are::

- ✓ Ubuntu Server
- ✓ Windows Server
- ✓ Red Hat Enterprise

Popular mobile OSs are::

- ✓ iPhone OS
- ✓ Android OS
- ✓ Windows Phone OS

Some concepts in OS

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Single Programming OS:

- Only one application program runs on top of OS.
- Single user operating system is that in which one user works on one interface.
- In these systems only one user is active at a time.
- These systems use less resources and they have less complexity so it is easy to maintain and debug.
- The system has a less chance to get damage , since number of requests from hardware and software is few.

This is Single-User , Single-Tasking OS.

-- e.g MS-DOS

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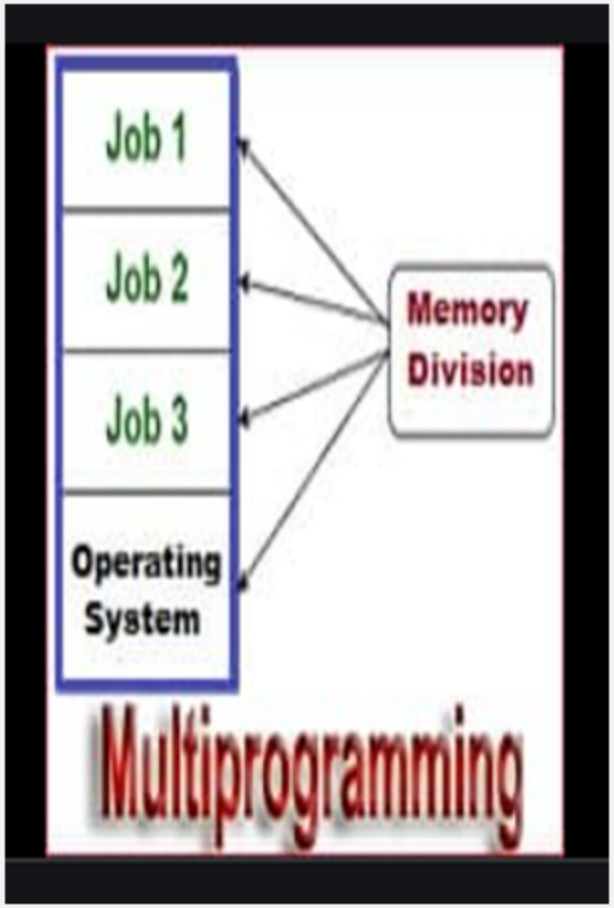
Disadvantage of Single Programming System:

- Major disadvantage of this system is that, it is very slow and it fails to handle work load efficiently.
- This OS has lot of limits and also CPU utilization is very poor , since CPU has to wait for another program to be brought into the memory after completion of execution of first program.
- Hence the concept of Multiplogramming evolved in Operating System.

Multiprogramming

- In MultiProgramming, more than one program gets loaded in memory along with OS.
- CPU remains busy always.
- Processor switches rapidly between the programs available in the memory.
- In this case , OS starts executing next job when--
  1. Program terminates
  2. Program is waiting for i/o services.
- This is non-preemptive system. CPU will process till the job is completed fully. CPU will switch to other job in case if the job goes for i/o functions or job terminates logically.
- System actually processess only one program at a time. The switching between programs is very fast .



-e.g Batch Processing system.

Multiprocessing/Time Sharing OS/Multitasking

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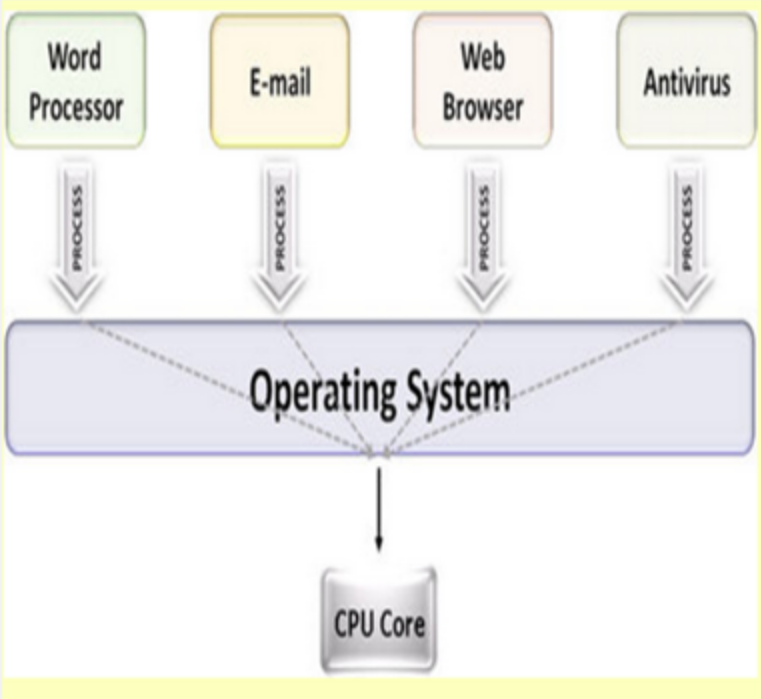
- Idea is to process multiple program simulteniously.
- Time slice/quantum is defined by OS.
- In this case OS will switch to other job when-----

✓ Time slice/quantum ends.

✓ Program is waiting for i/o operations to complete.

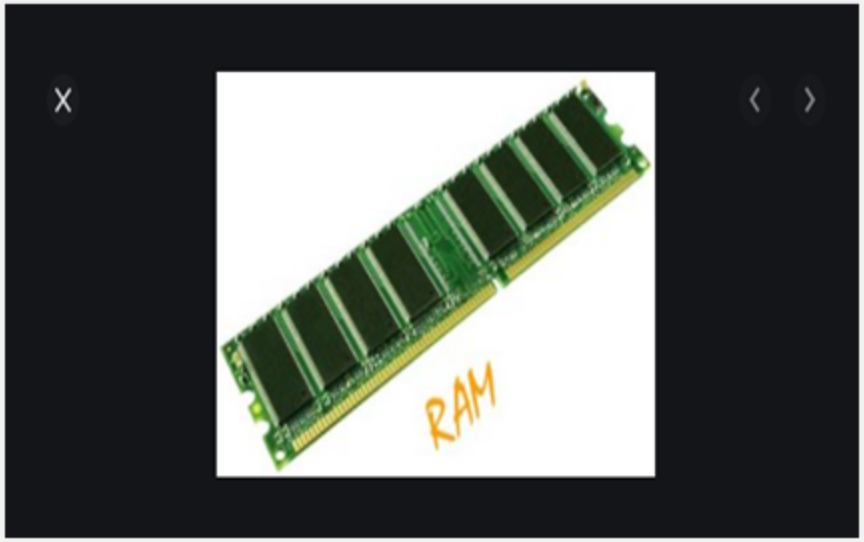
✓ Program terminates.

- Multiple programs are brought into the memory.
- In ideal case, CPU will process each program for definite time quantum and then switch to the other program.
- This switching is very fast and though only one program is executed at any point of time, the user gets the effect, that multiple programs are running in parallel.
- The CPU is never idle and is utilized to its optimum capacity.



- This increases speed and throughput of the system.
- This type of a system requires extra h/w called as a Timer.

Memory



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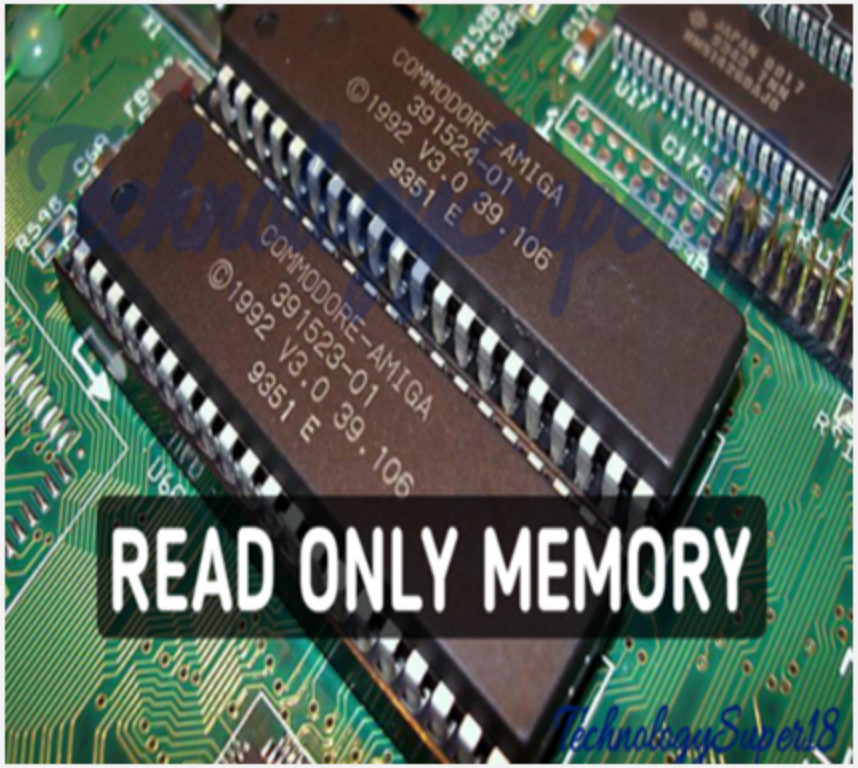
- RAM(Random Access Memory)
- RAM is also called as main/base memory.

Temporary Memory

- This type of memory , as the name says is temporary.
- The data in this memory will be cleared if system shuts down or there is a power-cut.
- Since the memory is erased due to power failure , it is called as Volatile memory.
- All the processing and loading of OS is done in this memory.
- CPU can directly access this memory.

Permanent Memory

- Memory is permanent.
- Data stored stays permanent and needs to be deleted explicitly .
- If the data is to be processed from this memory, it is brought into RAM for processing.

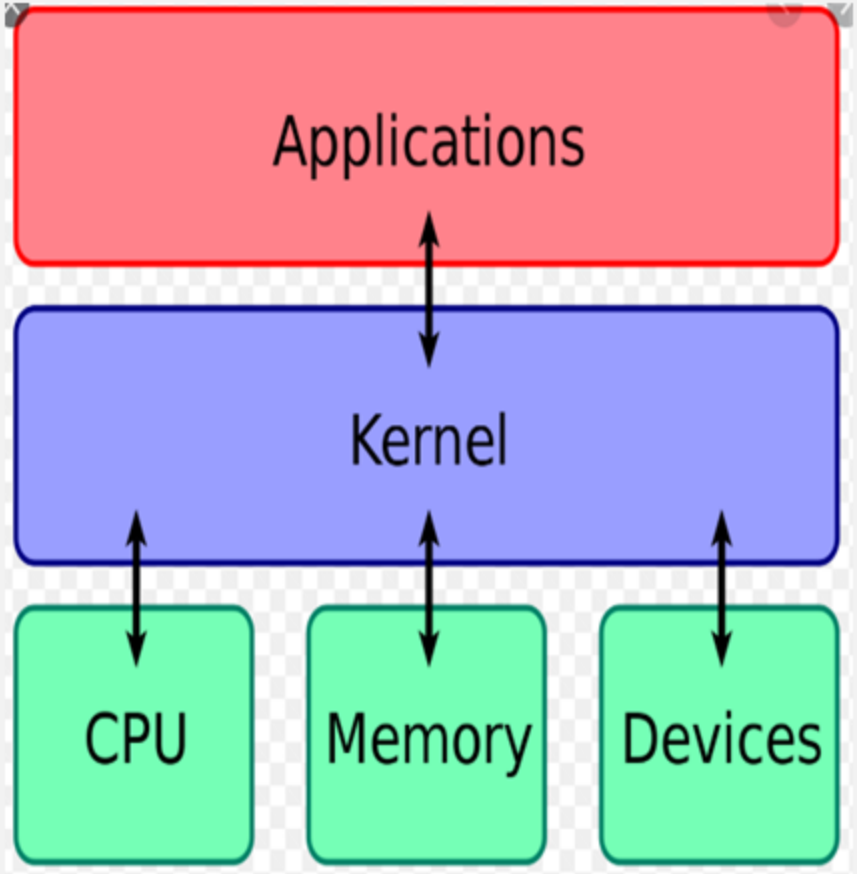


eg.

- ROM (Read Only Memory).
-All important programs required to start the computer system are stored in ROM.

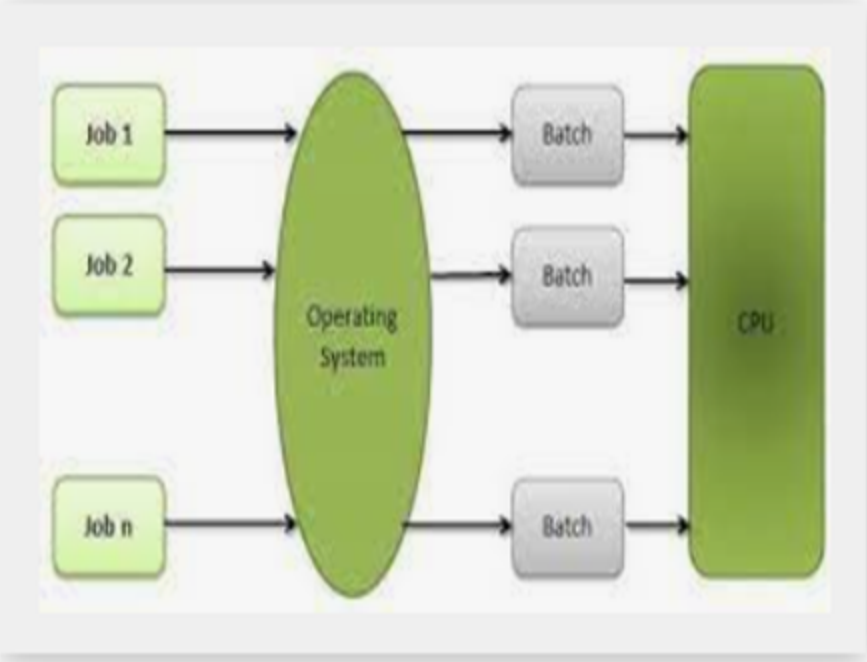
Kernel

- Kernel is said to be the "Heart of OS".
- A kernel is the core component of an operating system which performs all major tasks which we classify as the functions of Operating System.
- System calls are handled by Kernel.
- It is the first program to get loaded in memory.
- It runs till system shuts down. It talks to h/w of the system and most important , it " Handels interrupts ".



Batch Operating Syatem

- No interaction between user and the computer.
- The user submits a job (written on cards or tape) to computer operator.
- Computer operator then places a batch of several jobs on an input device.
- Jobs are batched together by type of languages and requirement.
- Multiple users can share the batch systems.



- Advantages::

It is easy to manage large work repeatedly done.

- Disadvantages::

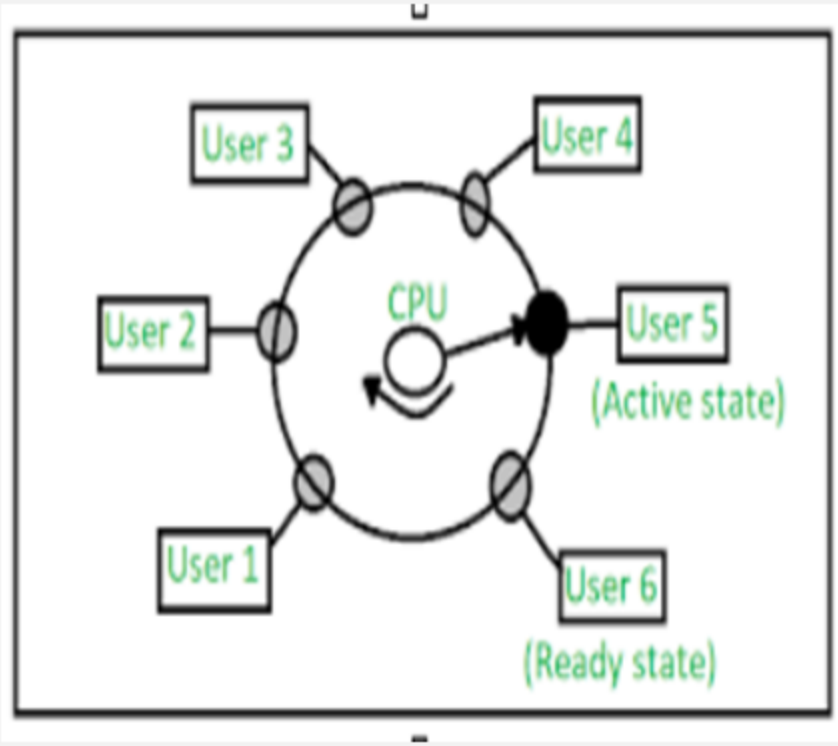
The computer operators should be well known with batch systems.
Batch systems are hard to debug.
It is sometime costly.
Some times the idle time in batch system is very high due to external machine speed .
The other jobs will have to wait for an unknown time if any job in the batch fails.
It is very difficult to guess or know the time required by any job to complete.

Examples ::

Payroll System,
Bank Statements etc.

Time-sharing Operating System

- A time sharing system allows many users to share the computer resources simultaneously.
- The time sharing system provides the direct access to a large number of users where CPU time is divided among all the users on scheduled basis.
- The OS allocates a set of time to each user called as time slice/time quantam.
- When this time is expired, it passes control to the next user on the system. The time allowed is extremely small and the user gets the impression that they each have their own CPU and they are the sole owner of the CPU.



Advantages ::

-Each task gets an equal opportunity.

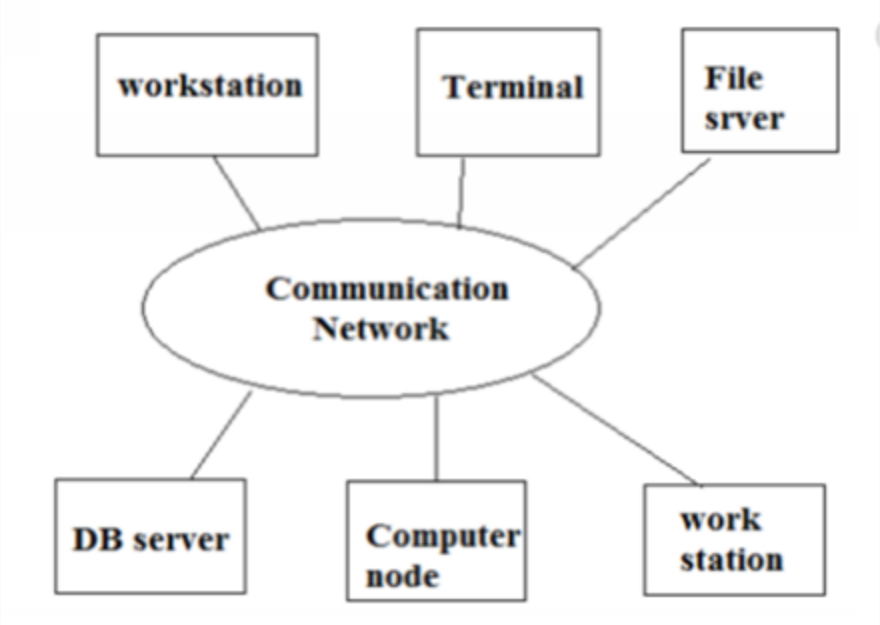
-CPU idle time can be reduced.

-Multiple programs can be executed.

Disadvantages ::

-One must take care of security and integrity of user programs and data.

Distributed Operating System



- In distributed OS data is stored and processed at multiple locations.
- Distributed systems comprises of many computers that are inter connected by communication networks.

Advantage::

--With resource sharing facility user at one site may be able to use the resources available at another site.

-- Speedy exchange of data with one another via electronic mail.

-- If one site fails in a distributed system, the remaining sites can potentially continue operating.

--Better service to the customers.

Disadvantage::

--Security problem due to sharing.

--Some messages can be lost in the network system.

--Bandwidth is another problem if there is large data then all network wires to be replaced which tends to become expensive.

Real-time systems

- These systems are used when time requirements are very strict like missile systems, air traffic control systems, robots etc.

- Responce time of these systems is very critical.

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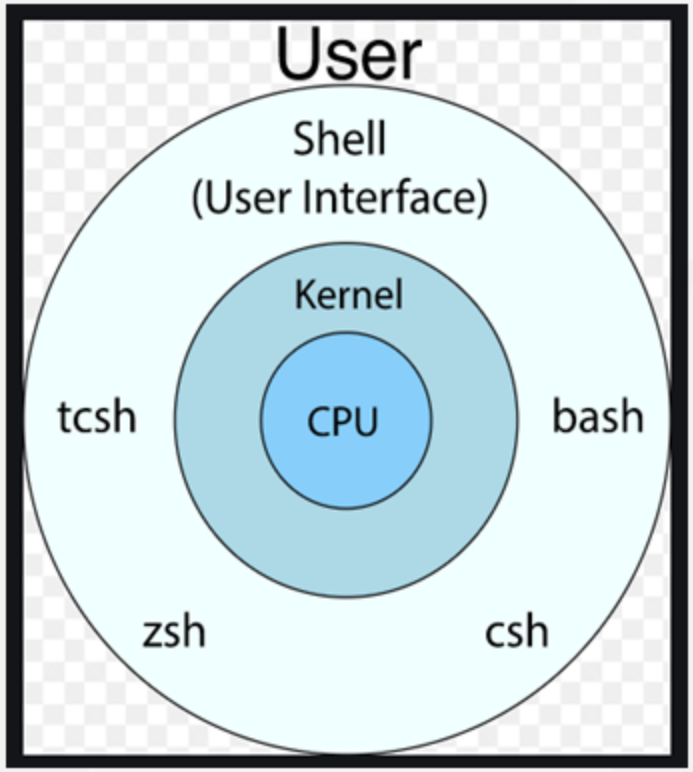
--Railway reservation system, Satellite tracking system.

Shell



- Interface between a user and an operating system service. Shell accepts human-readable commands.
- These commands are pre-compiled and programmed .
- Shell is command interpreter of Linux.
- Command Interpreter , reads and executes the command given by user.

Types of shell::

bourn shell – sh
bourn again shell – bash
korn shell – ksh

We use bash shell.
$ is our prompt.
Linux /mac OS use default shell as bash shell.
Unix has default shell as bourn shell.

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